

FRAUD CLAIM DETECTION

ML MODEL CASE STUDY

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PROBLEM STATEMENT

- To develop a ML model to help insurance company to identify the fraudulent claims to minimize financial losses .
- To optimize overall claims handling process to expedite approval process.



GOALS AND OBJECTIVES

- To build a model to classify insurance claims as either fraudulent or legitimate based on historical claim details and customer profiles. By using features like claim amounts, customer profiles and claim types, the company aims to predict which claims are likely to be fraudulent before they are approved.



QUESTIONS

- How can we analyse historical claim data to detect patterns that indicate fraudulent claims?
- Which features are the most predictive of fraudulent behaviour?
- Based on past data, can we predict the likelihood of fraud for an incoming claim?
- What insights can be drawn from the model that can help in improving the fraud detection process?



HOW CAN WE ANALYSE HISTORICAL CLAIM DATA TO DETECT PATTERNS THAT INDICATE FRAUDULENT CLAIMS?

Historical claim data can be analyzed using:

- **Exploratory Data Analysis (EDA)** to identify suspicious trends such as higher claim amounts, specific customer age groups, or certain hobbies linked with fraud.
- **Statistical summaries and visualizations** like boxplots, countplots, and correlation heatmaps to highlight patterns of outliers or inconsistencies.
- **Segmentation** based on variables like incident type, claim severity, and customer tenure to observe how fraud is distributed across groups.
- **Target Likelihood Encoding** to measure the proportion of fraud within each category of a feature.
- **Machine learning models** that assign weights to features based on their predictive power, helping us uncover hidden patterns not obvious from EDA alone.



WHICH FEATURES ARE THE MOST PREDICTIVE OF FRAUDULENT BEHAVIOUR?

From the Logistic Regression model and feature analysis:

- insured_hobbies_chess and insured_hobbies_cross-fit have strong positive coefficients, meaning claims associated with these hobbies are more likely to be fraudulent.
- incident_severity_Minor Damage and incident_severity_Total Loss are negatively associated with fraud, suggesting such claims are less suspicious.

Other predictive features include:

- months_as_customer — Newer customers showed a higher fraud rate.
- age — Younger individuals (under 30) were more fraud-prone.
- number_of_vehicles_involved and bodily_injuries — Higher counts often correlated with fraud.
- policy_deductible — Lower deductible amounts appeared to attract more fraudulent claims.



BASED ON PAST DATA, CAN WE PREDICT THE LIKELIHOOD OF FRAUD FOR AN INCOMING CLAIM?

Yes, based on historical data, the model can accurately predict the likelihood of fraud for a new claim.

- The Logistic Regression model achieved a Recall of 90.5% on validation data, meaning it is highly effective at identifying fraudulent claims.
- The model was trained using labelled past claims, learning patterns and signals commonly associated with fraud.
- With these patterns, the model assigns a probability score to each incoming claim, indicating how likely it is to be fraudulent.
- This enables proactive fraud detection before claims are approved.



WHAT INSIGHTS CAN BE DRAWN FROM THE MODEL THAT CAN HELP IN IMPROVING THE FRAUD DETECTION PROCESS?

- **Flag high-risk profiles:** Claims with hobbies like chess or cross-fit should be flagged for deeper review.
- **Fast-track low-risk claims:** Minor damage and total loss claims are rarely fraudulent — these can be expedited.
- **Priorities ambiguous cases:** Claims that are neither minor nor total loss deserve more attention, as they tend to fall in a “grey zone” of fraud risk.
- **Watch new customers:** Claims from recently onboarded customers showed a higher fraud rate — early lifecycle fraud checks can be enforced.
- **Policy deductible patterns:** Claims with low deductibles had higher fraud rates, which can inform policy restructuring or risk-based pricing.
- These insights help the company move from **reactive detection** to a **proactive prevention strategy**, optimizing both resource allocation and customer experience.



APPROACH AND METHODOLOGY

- Data Preparation (EDA) – Finding missing values and removing the column which is having $>40\%$ missing values. Imputing '?' with level 'Not Provided'. Data to be converted to a suitable format. Perform Univariate, Bivariate and Multivariate Analysis.
- Model Building-Dummy variable creation, Scaling of variables and splitting in train test data. Variable selection with correct techniques and choosing best model out of it. Model evaluation using confusion metrics and compare given target.



MISSING VALUE TREATMENT

- There is a huge value of '?' in 3 columns. But removing the rows with '?' will cost us a lot of data and they are important columns. So, instead we are going to replace it with 'Not Provided'. This way we have all the data and almost no null values. In case these come up in the model, it will be of no use and we can drop it off then. We can remove them manually at the time of dummy creation.
- There are some columns that are unique identifier or completely empty. Hence, we should drop those as they are having very less predictive power.
- Fix data types of date column.



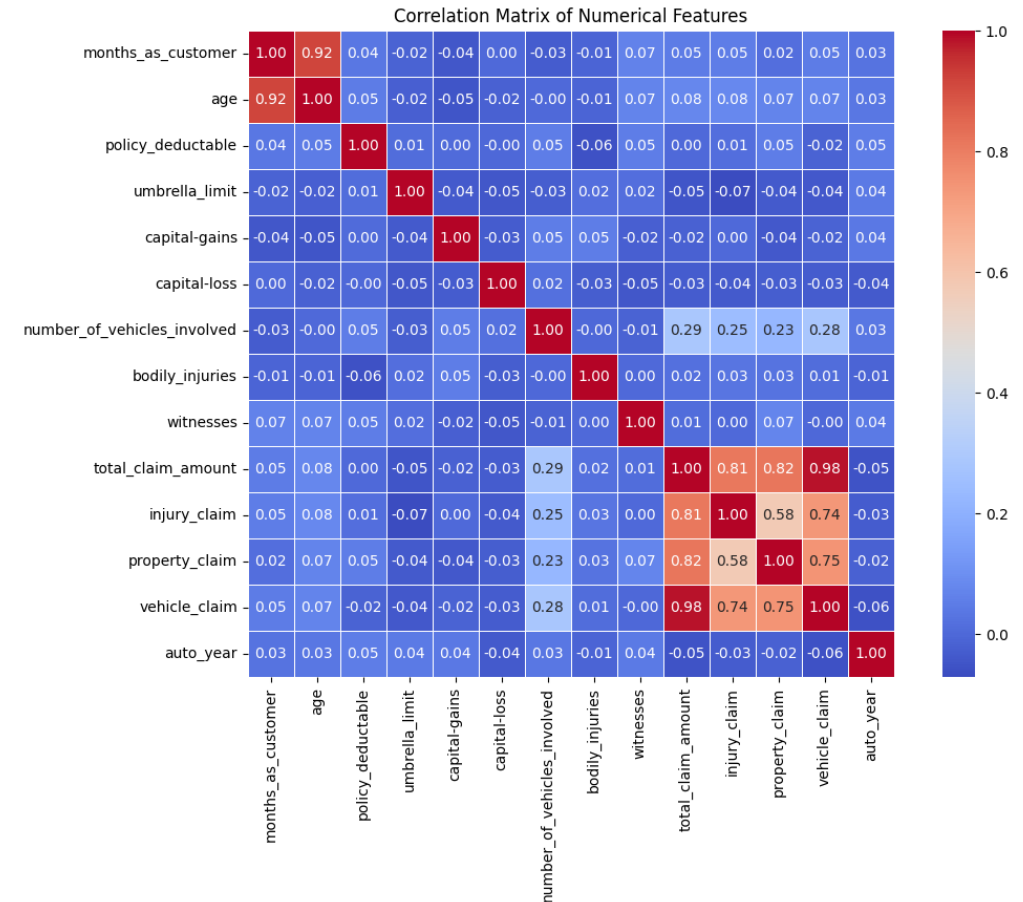
TRAIN TEST SPLIT

- Define and feature of target variable.
- The split was done at 70% and 30% for train and test data respectively.



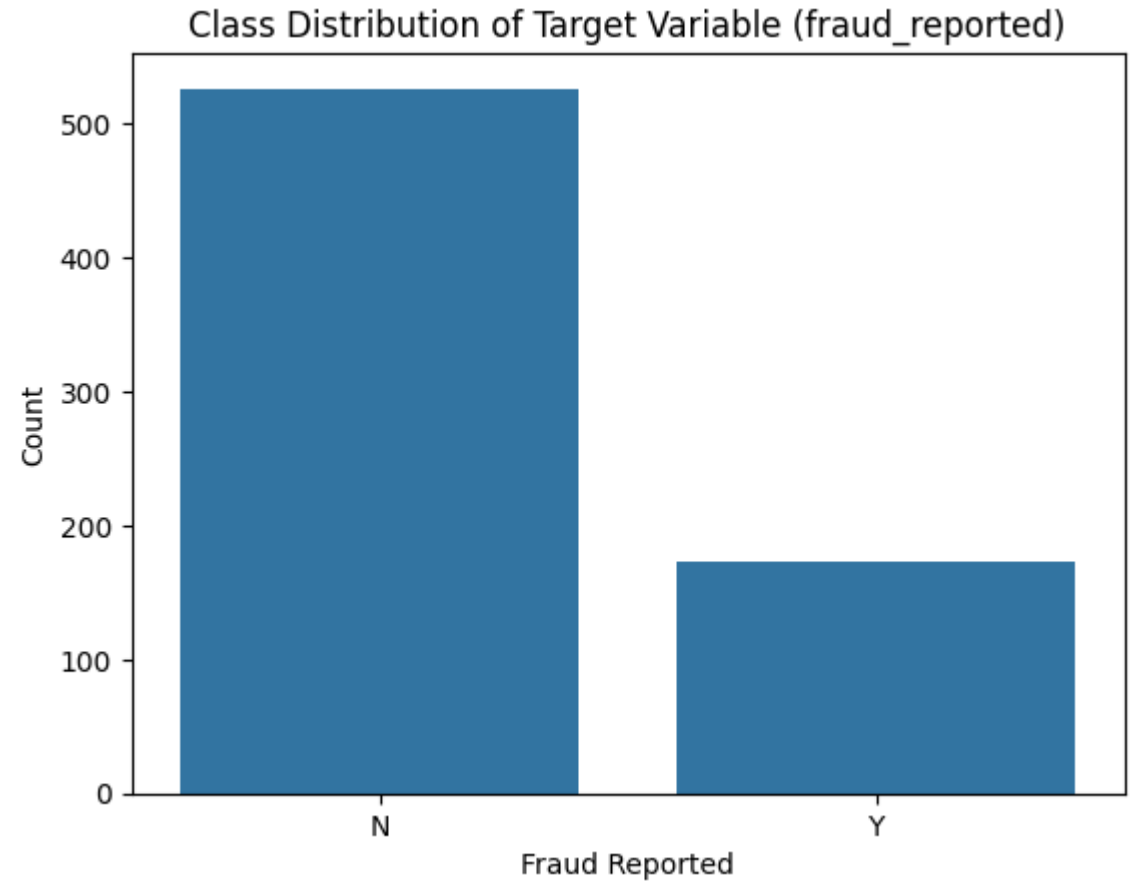
EDA

- Perform univariate, Bivariate and Multivariate Analysis.
- Perform Correlation Analysis.
- Check for highly co-related features.
- There aren't any major outliers.



CHECK CLASS BALANCE

- Check Target column for class balance.
- Here the target column is fraud_reported.
- Here class is highly imbalanced with 77% of Non-fraud and 23% of fraud



FEATURE ENGINEERING

- Perform resampling by using RandomOverSampler technique to balance the data and handle class imbalance.
- Feature Creation: Create new features from existing ones to enhance the model's ability to capture patterns in the data. Like by calculating the difference between incident_date and policy_bind_date we can know the no of days lapsed before incident.
- Handle redundant columns: Drop highly co-related columns .Also, Categorical columns with low value counts for some levels can be remapped to reduce number of unique levels, and features with very high counts for just one level may be removed, as they resemble unique identifier columns and do not provide substantial predictive value.
- Categorical variable with high no of category can be reclassified to reduced the levels.
- Binning can be done to the columns which is having numerical values but represented as categorical.
- Combine categories that have low frequency or provide limited predictive information
- Additionally, eliminate any columns from which the necessary features have already been extracted in the preceding step.



DUMMY VARIABLE CREATION AND SCALING

- The dummy variables were created for both train and test data where no of category>2 in categorical variables. Remove the dummy column manually where it is suffix is “Not Provided”
- For scaling train and test data on numeric values we used the MinMaxScaler.



MODEL BUILDING LOGISTIC REGRESSION

- RFECV done to attain the top relevant variables.
- Later the rest of the variables were removed manually depending on the VIF values and p-value.(variables with $VIF < 5$ and $p\text{-value} < 0.05$ to be considered)



MODEL BUILDING RANDOM FOREST

- Random forest without hyperparameter tuning `feature_importances_` is used and with hyperparameter tuning -GridSearchCV done to attain the best estimator.
- High important features have been selected and model fitting done on features selected.
- Model build by using hyperparameter tuning to enhance the performance. It uses `best_estimator_` to select its optimum feature.



RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

What the ROC Curve Shows:

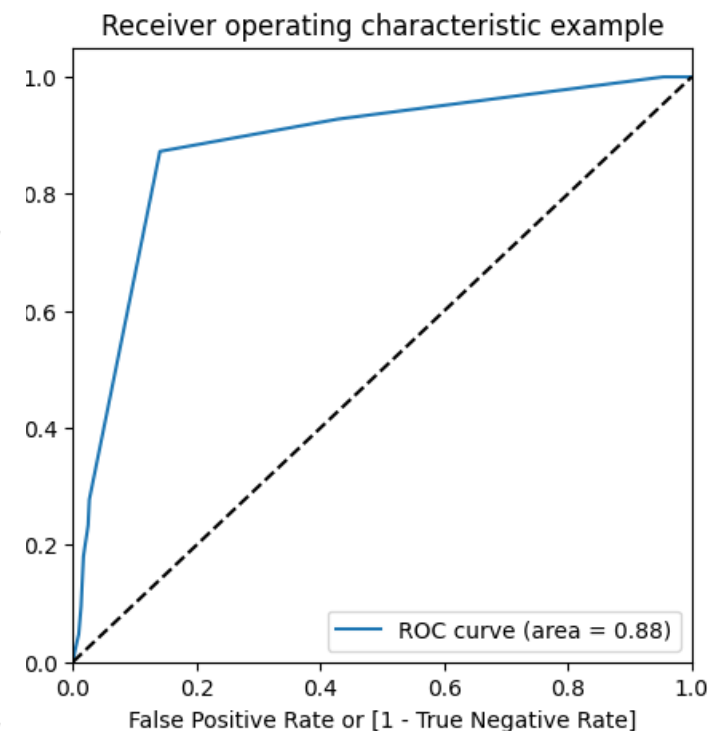
- The ROC curve plots the **True Positive Rate (Sensitivity)** against the **False Positive Rate (1 - Specificity)** at various classification thresholds.
- The **diagonal dashed line** represents a model with no discrimination power (i.e., a random guess with AUC = 0.5).
- The **blue curve** is our model's performance.

Area Under the Curve (AUC) = 0.88

- An **AUC of 0.88** indicates that the model has **excellent discrimination ability**.
- It means there is an **88% chance** that the model will correctly distinguish between a randomly chosen fraudulent and non-fraudulent claim.
- AUC ranges:
 - 0.90 – 1.0 : Excellent
 - 0.80 – 0.90 : **Good**
 - 0.70 – 0.80 : Fair
 - 0.50 – 0.70 : Poor

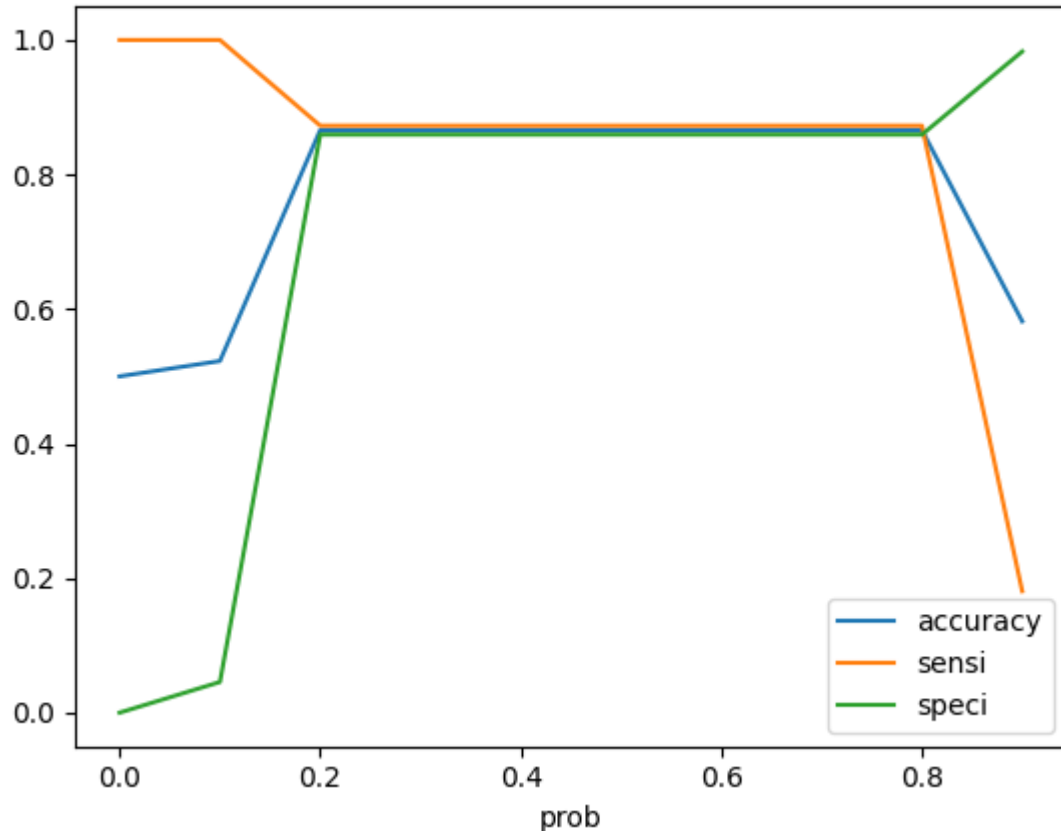
Conclusion:

- The model performs **well in separating fraud and non-fraud cases**.
- With an AUC of **0.88**, this model can be considered **reliable for classification**, though further tuning may still improve performance.



FINDING OPTIMAL CUT OFF VALUE- ACCURACY, SENSITIVITY, AND SPECIFICITY

Finding Optimal Cut Off value- accuracy, sensitivity, and specificity



The chart shows how changes in the probability threshold impact key performance metrics—accuracy (blue), sensitivity (orange), and specificity (green)—for the logistic regression model used in fraud detection. This visual helps in selecting a threshold that strikes the right balance between correctly identifying fraud and minimizing false positives.

We selected **0.4** as the cut-off value for the model, as thresholds between **0.2 and 0.8** produce similar prediction probabilities, making 0.4 a balanced and reliable choice.



LOGISTIC REGRESSION EVALUATION METRICS — TRAIN AND TEST DATASETS

Metric	Train dataset	Test dataset
Accuracy	86.6%	84.0%
Sensitivity	87.3%	90.5%
Specificity	85.9%	81.9%
Precision	86.1%	62.0%
Recall	87.3%	90.5%
F1-Score	86.7%	73.6%



RANDOM FORESTS EVALUATION METRICS — TRAIN AND TEST DATASETS

Metric	Train dataset	Test dataset
Accuracy	90.4%	77.7%
Sensitivity	93.3%	75.7%
Specificity	87.5%	78.3%
Precision	88.2%	53.3%
Recall	93.3%	75.7%
F1-Score	90.7%	62.6%



Evaluation and Conclusion

Insights

Based on the final model outputs, the following key insights were identified:

1. Hobby: Cross-Fit

- Claims filed by individuals who list **cross-fit** as a hobby show a strong positive association with fraud.
- A high coefficient (3.5445) indicates a greater likelihood of fraudulent behaviour in such cases.

2. Hobby: Chess

- Similarly, insured individuals who report **chess** as a hobby are also more likely to be associated with fraudulent claims.
- The model's coefficient (4.2153) suggests an increased log-odds of fraud for these claims.

3. Incident Severity: Minor Damage

- Claims categorized as **Minor Damage** are negatively associated with fraud.
- A significant negative coefficient (-3.4093) implies these claims are less likely to be fraudulent.

4. Incident Severity: Total Loss

- Claims involving **Total Loss** also show a negative association with fraud.
- The coefficient (-3.2046) suggests that such claims are relatively safer and less suspicious.



✓ Recommendations

1. Flag High-Risk Hobbies for Enhanced Scrutiny

- Implement additional verification checks for claims where the insured lists **chess** or **cross-fit** as hobbies.
- These categories have shown a higher correlation with fraud and should be prioritised for manual review.

2. Streamline Processing for Minor Damage Claims

- Since **minor damage** claims exhibit lower fraud probability, these can be **fast-tracked**.
- Automating their approval can improve efficiency and reduce turnaround time.

3. Optimise Resources on Total Loss Cases

- Given the low fraud risk associated with **total loss** claims, fewer manual checks may be needed.
- This allows resources to be redirected toward higher-risk claims.

4. Focus on Intermediate Severity Cases

- Claims that don't fall under either **Minor Damage** or **Total Loss** deserve closer attention.
- These cases may have a higher potential for fraudulent activity and should undergo **enhanced fraud detection** procedures.

✦ By combining data-driven insights with operational strategies, Global Insure can reduce fraudulent payouts and improve claim processing efficiency.

